Classical Field Theory — Exercise Sheet 11

Constraints and boundaries

Hints and answers

Use Lecture 11's canonical bracket, with $\mathbf{\Pi} = -\mathbf{E}$. A primary constraint is imposed only after forming the Hamilton equations. All field data below are real and smooth. The second problem uses Schwartz data on $\mathbb{R}^3$; the third explicitly changes the spatial region and its allowed boundary variations.

Exercise 11.1 (An entire constraint chain). Consider two mechanical coordinates $q(t)$ and $u(t)$ with Lagrangian

$$L(q, u, \dot{q}, \dot{u}) = \frac{1}{2}(\dot{q} - u)^2.$$

Use the following consistency procedure: find the image of the Legendre map, form the canonical Hamiltonian on that image, add an arbitrary multiplier for each primary constraint, and preserve the constraints until no new condition appears. The bracket on $(q, u; p_q, p_u)$ is the ordinary canonical bracket of mechanics.

(a) Carry out every step. State which conditions hold strongly and which hold only on the constraint surface. Classify the constraints.

(b) Find a time-dependent generator producing $\delta q = \epsilon(t)$ and $\delta u = \dot{\epsilon}(t)$ for an arbitrary smooth real function ϵ. Determine the transformation of the total-Hamiltonian multiplier needed to map solutions to solutions.

(c) At a fixed time, show that every two points on the constraint surface are gauge equivalent. How many canonical pairs remain on this quotient?

Hint. The relation for p_u contains no velocity. Preserve it before imposing it inside brackets. The values of ϵ and $\dot{\epsilon}$ can be prescribed independently at one time.

Answer. The constraints are $p_u = p_q = 0$, both first class. The total Hamiltonian is $p_q^2/2 + up_q + \lambda p_u$. The generator is $\epsilon p_q + \dot{\epsilon} p_u$ and $\lambda' = \lambda + \ddot{\epsilon}$. The quotient has no canonical pairs.

Exercise 11.2 (A mass term changes the constraint class). On $\mathbb{R}^{1+3}$ take the vacuum density

$$L_M = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 + \frac{M^2}{2}(\phi_{\mathrm{em}}^2 - |\mathbf{A}|^2), \qquad M > 0.$$

At each time all canonical fields and the multiplier are Schwartz. Use the extended bracket of Lecture 11 and define $P[\eta] = \int \eta\pi_\phi \, \mathrm{d}^3x$ for real Schwartz η.

(a) Derive the momenta, canonical Hamiltonian and consistency chain. Does preserving the secondary constraint leave $\partial_t\phi_{\mathrm{em}}$ arbitrary?

(b) Put $\mathcal{D}_M = \operatorname{div}\mathbf{\Pi} - M^2\phi_{\mathrm{em}}$ and $C_M[\chi] = -\int \chi\mathcal{D}_M \, \mathrm{d}^3x$. Compute all primary/secondary brackets. For smearings with $\int \eta\chi = 1$, write the two-by-two off-diagonal bracket block and its inverse. Why does its inverse have no massless limit?

(c) Impose both constraints and express the scalar variables in terms of $\mathbf{A}$ and $\mathbf{\Pi}$. Pull back the canonical form and determine how many spatial component pairs remain. Contrast this with Maxwell's gauge quotient.

Hint. The mass term changes neither the velocity Hessian nor $\mathbf{\Pi}$. It does change the scalar-potential derivative of the Hamiltonian. Keep the boundary term before moving $\nabla\phi_{\mathrm{em}}$ onto $\mathbf{\Pi}$.

Answer. The primary constraint is $\pi_\phi = 0$, and the secondary one is $\mathcal{D}_M = 0$. Its preservation fixes $\partial_t\phi_{\mathrm{em}} = -\operatorname{div}\mathbf{A}$. The mixed bracket is $\{P[\eta], C_M[\chi]\} = -M^2\int\eta\chi$. Three spatial pairs remain; the longitudinal pair is not a Maxwell gauge direction.

Exercise 11.3 (A boundary generator need not vanish). Let $\Omega = B_1(0) \subset \mathbb{R}^3$. For this problem the phase variables $\mathbf{A}, \mathbf{\Pi}$ are smooth on a neighbourhood of $\overline{\Omega}$, with no fixed normal trace for $\mathbf{\Pi}$. Their canonical form is $\int_\Omega \delta A_i \wedge \delta\Pi_i \, \mathrm{d}^3x$. A functional has a *smooth bulk derivative* if its variation is an integral of smooth coefficients against arbitrary smooth field variations, without a boundary evaluation term. For a fixed smooth real χ, set

$$C_\Omega[\chi] = -\int_\Omega \chi \operatorname{div}\mathbf{\Pi}\, \mathrm{d}^3x, \qquad G_\Omega[\chi] = \int_\Omega \mathbf{\Pi}\cdot\nabla\chi\, \mathrm{d}^3x.$$

(a) Derive their relation, including the surface integral, and vary it. Explain why $C_\Omega[\chi]$ generally lacks a smooth bulk derivative when χ is nonzero on the boundary, while $G_\Omega[\chi]$ has one.

(b) Find the Hamiltonian vector of $G_\Omega[\chi]$. On the Gauss surface $\operatorname{div}\mathbf{\Pi} = 0$, must its value and its contraction with the constrained canonical form vanish?

(c) Evaluate both functionals and their boundary difference for $\mathbf{\Pi} = e_1$ and $\chi = x^1$. Identify the boundary assumption that distinguishes this problem from the all-space integrations by parts in Lecture 11.

Hint. Use the outward normal on the sphere. A compact interior test first determines a candidate bulk gradient. A free normal boundary variation then tests whether that candidate represents the whole derivative.

Answer. The differentiable completion is $G_\Omega = C_\Omega + \int_{\partial\Omega}\chi\mathbf{\Pi}\cdot n\,\mathrm{d}S$. Its vector is $(\nabla\chi, 0)$. In the example $C_\Omega = 0$ and $G_\Omega = 4\pi/3$, with the same nonzero boundary charge.